

DSA Questions + Patterns

500 Curated Problems for FAANG & Top-Tier Interview Preparation

Arrays • Strings • Trees • Graphs • Dynamic Programming • Design

507

Problems parsed

13

Major sections

Direct

LeetCode links

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Modernized layout with direct LeetCode problem links, difficulty badges, topic summary, and quick-reference structure.

How to use this guide

This document is a comprehensive, formally curated collection of 500 Data Structures and Algorithms (DSA) questions designed for candidates preparing for technical interviews at top technology companies including Google, Amazon, Apple, Netflix, Meta (FAANG), Microsoft, and other tier-1 organisations. Each question is accompanied by its associated algorithmic pattern, enabling a pattern-first study approach that dramatically improves problem-solving speed and accuracy during interviews. Questions are organized by topic and sub-topic. Difficulty is annotated as [Easy], [Medium], or [Hard]. Master the patterns, not just the problems.

Easy	Medium	Hard	Total
84	294	129	507
Foundation	Core practice	Stretch problems	Parsed list

Topic map

Section	Topic	Source Q Range	Parsed Count
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Section 1: Arrays

Q1 - Q50 • 50 problems • 5 pattern groups

50
Problems

1.1 Two Pointers

Pattern: Two Pointers — use two indices moving toward/away from each other to reduce $O(n^2)$ to $O(n)$.

#	Problem	Difficulty	Link
1	Two Sum (sorted array)	Easy	Open
2	Three Sum	Medium	Open
3	Four Sum	Medium	Open
4	Container With Most Water	Medium	Open
5	Trapping Rain Water	Hard	Open
6	Remove Duplicates from Sorted Array	Easy	Open
7	Move Zeroes to End	Easy	Open
8	Sort Colors (Dutch National Flag)	Medium	Open
9	Minimum Size Subarray Sum	Medium	Open
10	Squares of a Sorted Array	Easy	Open

1.2 Sliding Window

Pattern: Sliding Window — maintain a window of elements; expand/shrink to satisfy a constraint.

#	Problem	Difficulty	Link
11	Longest Substring Without Repeating Characters	Medium	Open
12	Maximum Sum Subarray of Size K	Easy	Open
13	Fruit Into Baskets (at most 2 distinct)	Medium	Open
14	Minimum Window Substring	Hard	Open
15	Longest Subarray with Ones after Replacement	Medium	Open
16	Permutation in String	Medium	Open
17	Find All Anagrams in a String	Medium	Open
18	Substring with Concatenation of All Words	Hard	Open
19	Max Consecutive Ones III	Medium	Open
20	Number of Subarrays with Product Less Than K	Medium	Open

1.3 Prefix Sum / Difference Array

Pattern: Prefix Sum — precompute cumulative sums for $O(1)$ range queries.

#	Problem	Difficulty	Link
21	Subarray Sum Equals K	Medium	Open
22	Range Sum Query — Immutable	Easy	Open
23	Continuous Subarray Sum (multiple of k)	Medium	Open
24	Product of Array Except Self	Medium	Open

#	Problem	Difficulty	Link
25	Find Pivot Index	Easy	Open
26	Count Subarrays with Equal 0s and 1s	Medium	Open
27	Subarray Sums Divisible by K	Medium	Open
28	Minimum Operations to Reduce X to Zero	Medium	Open
29	Running Sum of 1D Array	Easy	Open
30	Random Pick with Weight	Medium	Open

1.4 Kadane's / Greedy on Arrays

Pattern: Kadane's Algorithm — track local and global maximum for maximum subarray problems.

#	Problem	Difficulty	Link
31	Maximum Subarray (Kadane's)	Medium	Open
32	Maximum Product Subarray	Medium	Open
33	Best Time to Buy and Sell Stock	Easy	Open
34	Best Time to Buy and Sell Stock II	Medium	Open
35	Jump Game	Medium	Open
36	Jump Game II (minimum jumps)	Medium	Open
37	Gas Station	Medium	Open
38	Candy Distribution	Hard	Open
39	Partition Labels	Medium	Open
40	Minimum Number of Arrows to Burst Balloons	Medium	Open

1.5 Sorting & Searching in Arrays

Pattern: Sorting + Binary Search — sort once, then binary-search for $O(n \log n)$ solutions.

#	Problem	Difficulty	Link
41	Merge Intervals	Medium	Open
42	Insert Interval	Medium	Open
43	Meeting Rooms II (min conference rooms)	Medium	Open
44	Non-overlapping Intervals	Medium	Open
45	Largest Number (custom sort)	Medium	Open
46	Kth Largest Element in an Array	Medium	Open
47	Find Minimum in Rotated Sorted Array	Medium	Open
48	Search in Rotated Sorted Array	Medium	Open
49	Count of Smaller Numbers After Self	Hard	Open
50	Find the Duplicate Number	Medium	Open

Section 2: Strings

Q51 - Q90 • 38 problems • 4 pattern groups

38
Problems

2.1 String Manipulation & Hashing

Pattern: HashMap / Frequency Count — count character frequencies to compare or find patterns.

#	Problem	Difficulty	Link
51	Valid Anagram	Easy	Open
52	Group Anagrams	Medium	Open
53	Longest Common Prefix	Easy	Open
54	Reverse Words in a String	Medium	Open
55	String to Integer (atoi)	Medium	Open
56	Roman to Integer	Easy	Open
57	Integer to Roman	Medium	Open
58	Count and Say	Medium	Open
59	ZigZag Conversion	Medium	Open
60	Multiply Strings	Medium	Open
61	Longest Palindrome (by rearrangement)	Easy	Open
62	First Unique Character in a String	Easy	Open

2.2 Sliding Window on Strings

Pattern: Variable-size Sliding Window — track character counts inside a window.

#	Problem	Difficulty	Link
63	Longest Repeating Character Replacement	Medium	Open
64	Minimum Window Substring (revisited with string focus)	Hard	Open
65	Longest Substring with At Most K Distinct Characters	Medium	Open
66	Longest Substring with At Most Two Distinct Characters	Medium	Open
67	Longest Palindromic Substring	Medium	Open
68	Palindromic Substrings (count all)	Medium	Open

2.3 Pattern Matching

Pattern: KMP / Rabin-Karp / Z-Algorithm — efficient $O(n+m)$ substring search.

#	Problem	Difficulty	Link
69	Implement strStr() / Needle in Haystack	Easy	Open
70	Repeated Substring Pattern	Easy	Open
71	Shortest Palindrome (KMP)	Hard	Open
72	Longest Happy Prefix (KMP failure function)	Hard	Open
73	Find the Index of the First Occurrence in a String	Easy	Open
74	Wildcard Matching	Hard	Open

#	Problem	Difficulty	Link
75	Regular Expression Matching	Hard	Open
76	Word Break	Medium	Open
77	Word Break II	Hard	Open
78	Decode Ways	Medium	Open

2.4 Stack-based String Problems

Pattern: Monotonic Stack / Explicit Stack — process characters left-to-right with a stack.

#	Problem	Difficulty	Link
79	Valid Parentheses	Easy	Open
80	Minimum Remove to Make Valid Parentheses	Medium	Open
81	Score of Parentheses	Medium	Open
82	Decode String (k[encoded_string])	Medium	Open
83	Remove All Adjacent Duplicates in String	Easy	Open
84	Remove K Digits	Medium	Open
85	Basic Calculator II	Medium	Open
86	Evaluate Reverse Polish Notation	Medium	Open
87	Remove Duplicate Letters (lexicographically smallest)	Medium	Open
88	Largest Rectangle in Histogram (string of bars)	Hard	Open

Section 3: Linked Lists

Q91 – Q120 • 30 problems • 3 pattern groups

30
Problems

3.1 Fast & Slow Pointers

Pattern: Floyd's Cycle Detection — two pointers at different speeds to detect cycles and midpoints.

#	Problem	Difficulty	Link
89	Linked List Cycle Detection	Easy	Open
90	Linked List Cycle II (entry point)	Medium	Open
91	Find the Middle of Linked List	Easy	Open
92	Happy Number (cycle in sequence)	Easy	Open
93	Palindrome Linked List	Easy	Open
94	Reorder List ($L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_{n-1}$)	Medium	Open

3.2 Reversal & Merging

Pattern: In-place Reversal — reverse sublists or entire lists using pointer manipulation.

#	Problem	Difficulty	Link
95	Reverse a Linked List	Easy	Open
96	Reverse Linked List II (sub-list)	Medium	Open
97	Reverse Nodes in k-Group	Hard	Open
98	Rotate List by k	Medium	Open
99	Merge Two Sorted Lists	Easy	Open
100	Merge K Sorted Lists	Hard	Open
101	Sort List (merge sort on LL)	Medium	Open
102	Partition List around value x	Medium	Open
103	Remove Nth Node from End	Medium	Open
104	Delete Node in a Linked List (no head)	Easy	Open
105	Odd Even Linked List	Medium	Open
106	Intersection of Two Linked Lists	Easy	Open
107	Flatten a Multilevel Doubly Linked List	Medium	Open
108	Copy List with Random Pointer	Medium	Open
109	LRU Cache (design problem)	Medium	Open

3.3 Advanced Linked List

#	Problem	Difficulty	Link
110	LFU Cache	Hard	Open
111	Design Skiplist	Hard	Open
112	Add Two Numbers (LL representation)	Medium	Open
113	Add Two Numbers II (no reversal)	Medium	Open
114	Swap Nodes in Pairs	Medium	Open

#	Problem	Difficulty	Link
115	Remove Duplicates from Sorted List II	Medium	Open
116	Next Greater Node in Linked List	Medium	Open
117	Insert into a Sorted Circular Linked List	Medium	Open
118	Convert Binary Number in LL to Integer	Easy	Open

Section 4: Stacks & Queues

Q121 - Q145 • 25 problems • 2 pattern groups

25
Problems

4.1 Monotonic Stack

Pattern: Monotonic Stack — maintain a stack in increasing/decreasing order for next-greater/smaller

#	Problem	Difficulty	Link
119	Next Greater Element I	Easy	Open
120	Next Greater Element II (circular)	Medium	Open
121	Daily Temperatures	Medium	Open
122	Largest Rectangle in Histogram	Hard	Open
123	Maximal Rectangle in Binary Matrix	Hard	Open
124	Sum of Subarray Minimums	Medium	Open
125	Maximum Width Ramp	Medium	Open
126	Online Stock Span	Medium	Open
127	132 Pattern	Medium	Open
128	Asteroid Collision	Medium	Open

4.2 Queue / Deque Patterns

Pattern: Monotonic Deque — maintain a deque for sliding window max/min in $O(n)$.

#	Problem	Difficulty	Link
129	Sliding Window Maximum	Hard	Open
130	First Negative in Every Window of Size K	Medium	Open
131	Implement Stack using Queues	Easy	Open
132	Implement Queue using Stacks	Easy	Open
133	Design Circular Queue	Medium	Open
134	Design Circular Deque	Medium	Open
135	Max of Min for Every Window Size	Hard	Open
136	Shortest Subarray with Sum at Least K	Hard	Open
137	Jump Game VI (DP + deque)	Medium	Open
138	Constrained Subsequence Sum	Hard	Open
139	Task Scheduler	Medium	Open
140	Number of Recent Calls	Easy	Open
141	Dota2 Senate	Medium	Open
142	Reveal Cards in Increasing Order	Medium	Open
143	Rotten Oranges (BFS)	Medium	Open

Section 5: Binary Search

Q146 - Q175 • 31 problems • 3 pattern groups

31
Problems

5.1 Classic Binary Search

Pattern: Binary Search — repeatedly halve the search space on a monotonic function.

#	Problem	Difficulty	Link
144	Binary Search (basic)	Easy	Open
145	First Bad Version	Easy	Open
146	Search Insert Position	Easy	Open
147	Count of Range Sum (advanced)	Hard	Open
148	Find Peak Element	Medium	Open
149	Find Minimum in Rotated Sorted Array II (duplicates)	Hard	Open
150	Search a 2D Matrix	Medium	Open
151	Search a 2D Matrix II	Medium	Open
152	Kth Smallest Element in Sorted Matrix	Medium	Open
153	Find K Closest Elements	Medium	Open

5.2 Binary Search on Answer

Pattern: Parametric Search — binary search on the answer space, validate with a feasibility function.

#	Problem	Difficulty	Link
154	Capacity to Ship Packages Within D Days	Medium	Open
155	Split Array Largest Sum	Hard	Open
156	Koko Eating Bananas	Medium	Open
157	Minimum Number of Days to Make m Bouquets	Medium	Open
158	Magnetic Force Between Two Balls	Medium	Open
159	Find the Smallest Divisor Given a Threshold	Medium	Open
160	Allocate Minimum Number of Pages	Medium	Open
161	Aggressive Cows (SPOJ classic)	Medium	Open
162	Painters Partition Problem	Hard	Open
163	EKO — Cutting Trees (binary search on height)	Medium	Open
164	Median of Two Sorted Arrays	Hard	Open
165	K-th Smallest Prime Fraction	Hard	Open
166	Find K-th Smallest Pair Distance	Hard	Open
167	Count of Pairs with Sum Less Than Target	Easy	Open
168	Maximum Running Time of N Computers	Hard	Open

5.3 Bisect Tricks

#	Problem	Difficulty	Link
169	Longest Increasing Subsequence ($O(n \log n)$)	Medium	Open

#	Problem	Difficulty	Link
170	Russian Doll Envelopes	Hard	Open
171	Count of Smaller Numbers After Self (BIT/BST)	Hard	Open
172	Minimum Operations to Make Array Increasing	Easy	Open
173	Maximum Profit in Job Scheduling	Hard	Open
174	Number of Longest Increasing Subsequences	Medium	Open

Section 6: Trees

Q176 - Q230 • 56 problems • 5 pattern groups

56
Problems

6.1 Tree Traversals

Pattern: DFS (Pre/In/Post-order) & BFS (Level-order) — foundation of nearly all tree problems.

#	Problem	Difficulty	Link
175	Binary Tree Inorder Traversal (iterative)	Easy	Open
176	Binary Tree Preorder Traversal (iterative)	Easy	Open
177	Binary Tree Postorder Traversal (iterative)	Easy	Open
178	Binary Tree Level Order Traversal	Medium	Open
179	Binary Tree Zigzag Level Order	Medium	Open
180	Binary Tree Right Side View	Medium	Open
181	Average of Levels in Binary Tree	Easy	Open
182	N-ary Tree Level Order Traversal	Medium	Open
183	Vertical Order Traversal	Hard	Open
184	Boundary Traversal of Binary Tree	Medium	Open

6.2 Binary Tree Properties

Pattern: Tree Recursion — solve sub-problems on left/right subtrees and combine results.

#	Problem	Difficulty	Link
185	Maximum Depth of Binary Tree	Easy	Open
186	Minimum Depth of Binary Tree	Easy	Open
187	Diameter of Binary Tree	Easy	Open
188	Balanced Binary Tree	Easy	Open
189	Same Tree	Easy	Open
190	Symmetric Tree	Easy	Open
191	Invert Binary Tree	Easy	Open
192	Count Complete Tree Nodes	Medium	Open
193	Sum of Left Leaves	Easy	Open
194	Path Sum	Easy	Open
195	Path Sum II (all paths)	Medium	Open
196	Binary Tree Maximum Path Sum	Hard	Open
197	Sum Root to Leaf Numbers	Medium	Open
198	Flatten Binary Tree to Linked List	Medium	Open
199	Populating Next Right Pointers	Medium	Open

6.3 Binary Search Trees (BST)

Pattern: BST Property — $\text{left} < \text{root} < \text{right}$; enables $O(\log n)$ search, insert, delete.

#	Problem	Difficulty	Link
200	Validate Binary Search Tree	Medium	Open
201	Kth Smallest Element in a BST	Medium	Open
202	Lowest Common Ancestor of BST	Easy	Open
203	Lowest Common Ancestor of Binary Tree	Medium	Open
204	Convert Sorted Array to BST	Easy	Open
205	Convert BST to Greater Tree	Medium	Open
206	Insert into a BST	Medium	Open
207	Delete Node in a BST	Medium	Open
208	Recover Binary Search Tree (two nodes swapped)	Hard	Open
209	Unique Binary Search Trees (count)	Medium	Open
210	Unique Binary Search Trees II (generate all)	Medium	Open
211	Serialize and Deserialize Binary Tree	Hard	Open
212	Binary Tree Cameras	Hard	Open
213	Construct BST from Preorder Traversal	Medium	Open
214	Two Sum IV in BST	Easy	Open

6.4 Segment Trees & BITs (Advanced)

Pattern: Segment Tree / Fenwick Tree — $O(\log n)$ range queries and point updates.

#	Problem	Difficulty	Link
215	Range Sum Query — Mutable (Segment Tree / BIT)	Medium	Open
216	Range Minimum Query	Medium	Open
217	Count of Range Sum (Merge Sort / BIT)	Hard	Open
218	Number of Longest Increasing Subsequences (Segment Tree)	Medium	Open
219	The Skyline Problem	Hard	Open
220	My Calendar I, II, III	Medium	Open
221	Rectangle Area II (coordinate compression)	Hard	Open
222	Falling Squares	Hard	Open
223	Interval List Intersections	Medium	Open
224	Data Stream as Disjoint Intervals	Hard	Open

6.5 Tries

Pattern: Trie (Prefix Tree) — efficient prefix search and autocomplete in $O(L)$ per query.

#	Problem	Difficulty	Link
225	Implement Trie (Prefix Tree)	Medium	Open
226	Word Search II (Trie + DFS)	Hard	Open
227	Design Add and Search Words Data Structure	Medium	Open
228	Replace Words with Root (Trie)	Medium	Open
229	Maximum XOR of Two Numbers (Trie)	Medium	Open
230	Palindrome Pairs (Trie)	Hard	Open

Section 7: Heaps & Priority Queues

Q232 - Q255 • 22 problems • 2 pattern groups

22
Problems

7.1 Top-K Pattern

Pattern: Min-Heap of size K — maintain the K largest elements efficiently.

#	Problem	Difficulty	Link
231	Kth Largest Element in a Stream	Easy	Open
232	Top K Frequent Elements	Medium	Open
233	Top K Frequent Words	Medium	Open
234	K Closest Points to Origin	Medium	Open
235	Find K Pairs with Smallest Sums	Medium	Open
236	Kth Smallest Element in Sorted Matrix (Heap)	Medium	Open
237	Sort Characters By Frequency	Medium	Open
238	Reorganize String	Medium	Open
239	Task Scheduler (Heap approach)	Medium	Open
240	Maximum Frequency Stack	Hard	Open

7.2 Two-Heap Pattern

Pattern: Two Heaps — one max-heap and one min-heap to find the median dynamically.

#	Problem	Difficulty	Link
241	Find Median from Data Stream	Hard	Open
242	Sliding Window Median	Hard	Open
243	IPO (Maximize Capital)	Hard	Open
244	Minimum Cost to Connect Sticks	Medium	Open
245	Smallest Range Covering Elements from K Lists	Hard	Open
246	Ugly Number II	Medium	Open
247	Super Ugly Number	Medium	Open
248	Meeting Rooms III (heap-based room assignment)	Hard	Open
249	Employee Free Time	Hard	Open
250	Process Tasks Using Servers	Medium	Open
251	Single-Threaded CPU	Medium	Open
252	Furthest Building You Can Reach	Medium	Open

Section 8: Graphs

Q256 - Q310 • 54 problems • 6 pattern groups

54
Problems

8.1 BFS (Shortest Path / Level-order)

Pattern: BFS — explore nodes level by level; guarantees shortest path in unweighted graphs.

#	Problem	Difficulty	Link
253	Number of Islands	Medium	Open
254	Max Area of Island	Medium	Open
255	Flood Fill	Easy	Open
256	Pacific Atlantic Water Flow	Medium	Open
257	01 Matrix (multi-source BFS)	Medium	Open
258	Rotting Oranges	Medium	Open
259	Walls and Gates (multi-source BFS)	Medium	Open
260	Shortest Path in Binary Matrix	Medium	Open
261	Snakes and Ladders	Medium	Open
262	Open the Lock	Medium	Open
263	Word Ladder	Hard	Open
264	Word Ladder II (all shortest paths)	Hard	Open
265	Minimum Knight Moves	Medium	Open
266	Bus Routes (BFS on routes)	Hard	Open
267	Cut Off Trees for Golf Event	Hard	Open

8.2 DFS / Backtracking on Graphs

Pattern: DFS with Visited Set — explore all paths; detect cycles and connected components.

#	Problem	Difficulty	Link
268	Clone Graph	Medium	Open
269	Course Schedule (cycle detection)	Medium	Open
270	Course Schedule II (topological sort)	Medium	Open
271	Number of Connected Components in Undirected Graph	Medium	Open
272	Graph Valid Tree	Medium	Open
273	Redundant Connection (Union-Find)	Medium	Open
274	Accounts Merge (Union-Find / DFS)	Medium	Open
275	All Paths from Source to Target	Medium	Open
276	Is Graph Bipartite?	Medium	Open
277	Possible Bipartition	Medium	Open

8.3 Topological Sort

Pattern: Kahn's Algorithm / DFS Post-order — linear ordering of nodes in a DAG.

#	Problem	Difficulty	Link
278	Alien Dictionary	Hard	Open
279	Sequence Reconstruction	Medium	Open
280	Minimum Height Trees	Medium	Open
281	Find Eventual Safe States	Medium	Open
282	Parallel Courses (min semesters)	Medium	Open
283	Sort Items by Groups Respecting Dependencies	Hard	Open

8.4 Union-Find (Disjoint Set Union)

Pattern: DSU with Path Compression + Union by Rank — near- $O(1)$ amortised operations.

#	Problem	Difficulty	Link
284	Find if Path Exists in Graph	Easy	Open
285	Number of Provinces	Medium	Open
286	Redundant Connection II (directed)	Hard	Open
287	Making a Large Island	Hard	Open
288	Swim in Rising Water	Hard	Open
289	Earliest Moment When Everyone Become Friends	Medium	Open
290	Satisfiability of Equality Equations	Medium	Open
291	Remove Max Number of Edges to Keep Graph Fully Traversable	Hard	Open

8.5 Shortest Path Algorithms

Pattern: Dijkstra / Bellman-Ford / Floyd-Warshall — weighted shortest paths.

#	Problem	Difficulty	Link
292	Network Delay Time (Dijkstra)	Medium	Open
293	Path with Maximum Probability	Medium	Open
294	Cheapest Flights Within K Stops (Bellman-Ford)	Medium	Open
295	Find the City with Smallest Number of Neighbors (Floyd-Warshall)	Medium	Open
296	Path with Minimum Effort (Dijkstra / Binary Search)	Medium	Open
297	Number of Ways to Arrive at Destination	Medium	Open
298	Minimum Weighted Subgraph with Required Paths	Hard	Open
299	Minimum Score of a Path Between Two Cities	Medium	Open
300	Reachable Nodes in Subdivided Graph	Hard	Open
301	K-th Shortest Path (Yen's Algorithm concept)	Hard	Open

8.6 Minimum Spanning Tree

Pattern: Kruskal's / Prim's — greedily build a spanning tree with minimum total edge weight.

#	Problem	Difficulty	Link
302	Min Cost to Connect All Points (Prim's)	Medium	Open
303	Optimize Water Distribution in a Village	Hard	Open
304	Critical Connections in a Network (Bridges)	Hard	Open

#	Problem	Difficulty	Link
305	Minimum Cost to Reach City with Tolls	Medium	Open
306	Find Critical and Pseudo-Critical Edges in MST	Hard	Open

Section 9: Dynamic Programming

Q311 – Q390 • 80 problems • 7 pattern groups

80

Problems

9.1 1-D DP (Linear Recurrences)

Pattern: 1-D DP — state is a single index; transition from previous states.

#	Problem	Difficulty	Link
307	Climbing Stairs	Easy	Open
308	House Robber	Medium	Open
309	House Robber II (circular)	Medium	Open
310	House Robber III (tree DP)	Medium	Open
311	Min Cost Climbing Stairs	Easy	Open
312	Fibonacci Number	Easy	Open
313	Tribonacci Number	Easy	Open
314	Coin Change (minimum coins)	Medium	Open
315	Coin Change II (number of ways)	Medium	Open
316	Perfect Squares	Medium	Open
317	Integer Break	Medium	Open
318	Decode Ways	Medium	Open
319	Decode Ways II	Hard	Open
320	Ugly Number III	Medium	Open
321	N-th Tribonacci Number	Easy	Open

9.2 2-D DP (Grid / Two Sequences)

Pattern: 2-D DP — state is two indices (i, j); fill a table row by row.

#	Problem	Difficulty	Link
322	Unique Paths	Medium	Open
323	Unique Paths II (with obstacles)	Medium	Open
324	Minimum Path Sum in Grid	Medium	Open
325	Dungeon Game (reverse DP)	Hard	Open
326	Maximal Square	Medium	Open
327	Count Square Submatrices with All Ones	Medium	Open
328	Longest Common Subsequence	Medium	Open
329	Shortest Common Supersequence	Hard	Open
330	Edit Distance	Hard	Open
331	Distinct Subsequences	Hard	Open
332	Interleaving String	Medium	Open
333	Regular Expression Matching (DP)	Hard	Open
334	Wildcard Matching (DP)	Hard	Open
335	Scramble String	Hard	Open

#	Problem	Difficulty	Link
336	Burst Balloons (interval DP)	Hard	Open

9.3 Knapsack Variants

Pattern: 0/1 Knapsack / Unbounded Knapsack — classic DP on items with weight/value.

#	Problem	Difficulty	Link
337	0/1 Knapsack Problem	Medium	Open
338	Unbounded Knapsack	Medium	Open
339	Partition Equal Subset Sum	Medium	Open
340	Target Sum (assign +/- to reach target)	Medium	Open
341	Last Stone Weight II (knapsack)	Medium	Open
342	Ones and Zeroes (2D knapsack)	Medium	Open
343	Profitable Schemes	Hard	Open
344	Number of Dice Rolls with Target Sum	Medium	Open
345	Combination Sum IV	Medium	Open
346	Count Ways to Build Good Strings	Medium	Open

9.4 Interval / Range DP

Pattern: Interval DP — solve subproblems over all intervals $[i, j]$, merge results.

#	Problem	Difficulty	Link
347	Matrix Chain Multiplication	Hard	Open
348	Minimum Cost Tree from Leaf Values	Medium	Open
349	Strange Printer	Hard	Open
350	Remove Boxes	Hard	Open
351	Zuma Game	Hard	Open
352	Minimum Cost to Merge Stones	Hard	Open
353	Optimal Strategy for a Game	Medium	Open
354	Palindrome Partitioning II (min cuts)	Hard	Open
355	Palindrome Partitioning IV (3 parts)	Hard	Open
356	Maximum Coins You Can Get	Medium	Open

9.5 State-Machine DP (Stocks & Transactions)

Pattern: State Machine DP — model allowed states (holding/not holding) and transitions.

#	Problem	Difficulty	Link
357	Best Time to Buy and Sell Stock III (at most 2 transactions)	Hard	Open
358	Best Time to Buy and Sell Stock IV (at most k transactions)	Hard	Open
359	Best Time to Buy and Sell Stock with Cooldown	Medium	Open
360	Best Time to Buy and Sell Stock with Transaction Fee	Medium	Open
361	Paint House	Medium	Open
362	Paint House II (k colors)	Hard	Open

#	Problem	Difficulty	Link
363	Paint Fence	Medium	Open
364	Student Attendance Record II	Hard	Open
365	Coin Path (min cost with k steps)	Hard	Open
366	Number of Ways to Stay in the Same Place After Some Steps	Hard	Open

9.6 Bitmask DP

Pattern: Bitmask DP — represent subsets as bitmasks; ideal for small n (≤ 20).

#	Problem	Difficulty	Link
367	Traveling Salesman Problem (bitmask DP)	Hard	Open
368	Partition to K Equal Sum Subsets	Medium	Open
369	Minimum XOR Sum of Two Arrays (assignment)	Hard	Open
370	Maximum Students Taking Exam	Hard	Open
371	Stickers to Spell Word	Hard	Open
372	Shortest Path Visiting All Nodes (BFS + bitmask)	Hard	Open
373	Number of Ways to Wear Different Hats to Each Other	Hard	Open
374	Count Ways to Distribute Candies	Hard	Open
375	Maximize Score After N Operations	Hard	Open
376	Find the Shortest Superstring (bitmask DP)	Hard	Open

9.7 DP on Trees & Graphs

#	Problem	Difficulty	Link
377	Diameter of N-ary Tree	Hard	Open
378	Binary Tree Maximum Path Sum (revisited)	Hard	Open
379	Maximum Sum of 3 Non-Overlapping Subarrays	Hard	Open
380	Cherry Pickup	Hard	Open
381	Cherry Pickup II (two robots)	Hard	Open
382	Minimum Difficulty of a Job Schedule	Hard	Open
383	Build Array Where You Can Find the Maximum Exactly K Comparisons	Hard	Open
384	Number of Music Playlists	Hard	Open
385	Count Vowels Permutation	Hard	Open
386	Minimum Cost to Cut a Stick	Hard	Open

Section 10: Backtracking

Q391 – Q420 • 30 problems • 3 pattern groups

30
Problems

10.1 Subsets & Combinations

Pattern: Backtracking — explore all choices recursively; prune branches that violate constraints.

#	Problem	Difficulty	Link
387	Subsets	Medium	Open
388	Subsets II (with duplicates)	Medium	Open
389	Combinations	Medium	Open
390	Combination Sum	Medium	Open
391	Combination Sum II (each number once)	Medium	Open
392	Combination Sum III (exactly k numbers)	Medium	Open
393	Letter Combinations of a Phone Number	Medium	Open
394	Generate Parentheses	Medium	Open
395	Count Numbers with Unique Digits	Medium	Open
396	Beautiful Arrangement	Medium	Open

10.2 Permutations

Pattern: Permutation Backtracking — swap elements or use a visited array.

#	Problem	Difficulty	Link
397	Permutations	Medium	Open
398	Permutations II (with duplicates)	Medium	Open
399	Next Permutation	Medium	Open
400	Permutation Sequence (kth permutation)	Hard	Open
401	Palindrome Partitioning	Medium	Open
402	Word Search	Medium	Open
403	N-Queens	Hard	Open
404	N-Queens II (count solutions)	Hard	Open
405	Sudoku Solver	Hard	Open
406	Remove Invalid Parentheses	Hard	Open

10.3 Advanced Backtracking

#	Problem	Difficulty	Link
407	Expression Add Operators	Hard	Open
408	Restore IP Addresses	Medium	Open
409	Letter Tile Possibilities	Medium	Open
410	Splitting a String Into Descending Consecutive Values	Medium	Open
411	Number of Squareful Arrays	Hard	Open
412	Maximum Length of a Concatenated String with Unique Characters	Medium	Open

#	Problem	Difficulty	Link
413	Tiling a Rectangle with the Fewest Squares	Hard	Open
414	Coloring a Border	Medium	Open
415	Factor Combinations	Medium	Open
416	Construct the Lexicographically Largest Valid Sequence	Medium	Open

Section 11: Greedy Algorithms

Q421 - Q445 • 25 problems • 2 pattern groups

25
Problems

11.1 Interval & Scheduling Greedy

Pattern: Greedy Selection — always pick the locally optimal choice; prove exchange argument.

#	Problem	Difficulty	Link
417	Activity Selection / Non-overlapping Intervals	Medium	Open
418	Minimum Number of Platforms Required	Medium	Open
419	Job Sequencing Problem	Medium	Open
420	Fractional Knapsack	Easy	Open
421	N meetings in one room	Easy	Open
422	Assign Cookies	Easy	Open
423	Lemonade Change	Easy	Open
424	Queue Reconstruction by Height	Medium	Open
425	Two City Scheduling	Medium	Open
426	Minimum Cost to Move Chips to Same Position	Easy	Open

11.2 String & Array Greedy

#	Problem	Difficulty	Link
427	Remove K Digits (smallest number)	Medium	Open
428	Largest Number After Removing k Digits (largest)	Medium	Open
429	Maximum Units on a Truck	Easy	Open
430	Minimum Deletions to Make Character Frequencies Unique	Medium	Open
431	Minimum Number of Flips to Make Binary String Alternating	Medium	Open
432	Minimum Swaps to Balance Parentheses	Medium	Open
433	Maximum Score from Removing Substrings	Medium	Open
434	Wiggle Subsequence	Medium	Open
435	Dota2 Senate (greedy queue)	Medium	Open
436	Minimum Time to Make Rope Colorful	Medium	Open
437	Boats to Save People	Medium	Open
438	Advantage Shuffle	Medium	Open
439	Maximum Performance of a Team	Hard	Open
440	Minimize Maximum Pair Sum in Array	Medium	Open
441	Earliest Deadline First Scheduling	Medium	Open

Section 12: Hashing, Math & Bit Manipulation

Q446 - Q470 • 30 problems • 3 pattern groups

30
Problems

12.1 HashMap / HashSet Patterns

Pattern: HashMap — $O(1)$ lookups to avoid nested loops; track frequency, index, or pairs.

#	Problem	Difficulty	Link
442	Two Sum (HashMap)	Easy	Open
443	Longest Consecutive Sequence	Medium	Open
444	4Sum II (count tuples)	Medium	Open
445	Isomorphic Strings	Easy	Open
446	Word Pattern	Easy	Open
447	Bulls and Cows	Medium	Open
448	Brick Wall (maximum bricks not cut)	Medium	Open
449	Subarray Sum Equals K (revisited with HashMap)	Medium	Open
450	Contiguous Array (equal 0s and 1s)	Medium	Open
451	Maximum Size Subarray Sum Equals k	Medium	Open

12.2 Math & Number Theory

Pattern: Number Theory — prime sieves, GCD/LCM, modular arithmetic, combinatorics.

#	Problem	Difficulty	Link
452	Count Primes (Sieve of Eratosthenes)	Medium	Open
453	Power of Two / Three / Four	Easy	Open
454	Excel Sheet Column Number	Easy	Open
455	Happy Number (Floyd's cycle)	Easy	Open
456	Ugly Number	Easy	Open
457	Reverse Integer	Medium	Open
458	Palindrome Number	Easy	Open
459	Factorial Trailing Zeroes	Medium	Open
460	Nth Digit	Medium	Open
461	Super Power (modular exponentiation)	Medium	Open

12.3 Bit Manipulation

Pattern: Bit Tricks — XOR, AND/OR masks, Brian Kernighan's algorithm for $O(1)$ bit operations.

#	Problem	Difficulty	Link
462	Single Number	Easy	Open
463	Single Number II (bit counting)	Medium	Open
464	Single Number III (two unique)	Medium	Open
465	Number of 1 Bits (Hamming Weight)	Easy	Open

#	Problem	Difficulty	Link
466	Counting Bits (0 to n)	Easy	Open
467	Reverse Bits	Easy	Open
468	Missing Number	Easy	Open
469	Sum of Two Integers (without + operator)	Medium	Open
470	Maximum XOR of Two Numbers in Array	Medium	Open
471	UTF-8 Validation	Medium	Open

Section 13: System Design & Data Structures Design

Q471 - Q500 • 36 problems • 4 pattern groups

36

Problems

13.1 Classic Design Problems

Pattern: OOP + Right Data Structures — combine hash maps, heaps, doubly linked lists for O(1) ops.

#	Problem	Difficulty	Link
472	Design LRU Cache	Medium	Open
473	Design LFU Cache	Hard	Open
474	Design Twitter (top 10 tweets feed)	Medium	Open
475	Design a Hit Counter	Medium	Open
476	Design a Rate Limiter (token bucket)	Medium	Open
477	Design In-Memory File System	Hard	Open
478	Design Search Autocomplete System (Trie + heap)	Hard	Open
479	Design a Phone Directory	Medium	Open
480	Implement Trie with Count	Medium	Open
481	Design Underground System (check-in/check-out)	Medium	Open

13.2 Randomized & Probabilistic Structures

Pattern: Reservoir Sampling / Fisher-Yates — uniform random sampling in O(1) space.

#	Problem	Difficulty	Link
482	Shuffle an Array	Medium	Open
483	Random Pick Index (reservoir sampling)	Medium	Open
484	Linked List Random Node	Medium	Open
485	Random Pick with Weight	Medium	Open
486	Insert Delete GetRandom O(1)	Medium	Open
487	Insert Delete GetRandom O(1) — Duplicates Allowed	Hard	Open

13.3 Stream / Online Algorithms

Pattern: Online Algorithms — process data one element at a time without looking back.

#	Problem	Difficulty	Link
488	Moving Average from Data Stream	Easy	Open
489	Find Median from Data Stream (revisited)	Hard	Open
490	Kth Largest Element in a Stream (revisited)	Easy	Open
491	Design a Stack with Increment Operation	Medium	Open
492	Design a Stack with getMin() / getMax()	Easy	Open
493	Time-Based Key-Value Store	Medium	Open
494	Design an Ordered Stream	Easy	Open
495	Snapshot Array	Medium	Open

#	Problem	Difficulty	Link
496	Design a Log Aggregation System	Medium	Open
497	First Unique Number in Stream	Medium	Open

13.4 Advanced Design Challenges

#	Problem	Difficulty	Link
498	Word Filter (prefix + suffix search, Trie)	Hard	Open
499	Exam Room (seat assignment)	Medium	Open
500	My Calendar I / II / III	Medium	Open
501	Range Module (add/remove/query ranges)	Hard	Open
502	All O'one Data Structure (O(1) min/max string freq)	Hard	Open
503	Design a Food Rating System	Medium	Open
504	Stock Price Fluctuation (max/min with updates)	Medium	Open
505	Find Servers That Handled Most Requests	Hard	Open
506	Booking Concert Tickets in Groups	Hard	Open
507	Subrectangle Queries	Medium	Open

Appendix A: Topic Summary & Pattern Quick-Reference

Section	Topic	Q Range	Key Pattern(s)
1	Arrays	1 – 50	Two Pointers, Sliding Window, Prefix Sum, Kadane's Algorithm, Sorting + Binary Search
2	Strings	51 – 90	HashMap / Frequency Count, Variable, KMP / Rabin, Monotonic Stack / Explicit Stack
3	Linked Lists	91 – 120	Floyd's Cycle Detection
4	Stacks & Queues	121 – 145	Monotonic Stack, Monotonic Deque
5	Binary Search	146 – 175	Binary Search, Parametric Search
6	Trees	176 – 230	DFS (Pre/In/Post, Tree Recursion, BST Property, Segment Tree / Fenwick Tree, Trie (Prefix Tree)
7	Heaps & Priority Queues	232 – 255	Min, Two Heaps
8	Graphs	256 – 310	BFS, DFS with Visited Set, Kahn's Algorithm / DFS Post, DSU with Path Compression + Union by Rank, Dijkstra / Bellman, Kruskal's / Prim's
9	Dynamic Programming	311 – 390	0/1 Knapsack / Unbounded Knapsack, Interval DP, State Machine DP, Bitmask DP
10	Backtracking	391 – 420	Backtracking, Permutation Backtracking
11	Greedy Algorithms	421 – 445	Greedy Selection
12	Hashing, Math & Bit Manipulation	446 – 470	HashMap, Number Theory, Bit Tricks
13	System Design & Data Structure Design	471 – 500	OOP + Right Data Structures, Reservoir Sampling / Fisher, Online Algorithms

Appendix B: Recommended 12-Week Study Plan

Week 1-2	Arrays (Q1-Q50) + Strings (Q51-Q90) - master sliding window and two-pointer patterns.
Week 3	Linked Lists (Q91-Q120) + Stacks & Queues (Q121-Q145) - pointer manipulation and monotonic structures.
Week 4	Binary Search (Q146-Q175) - classic and parametric search; practice writing bug-free lo/hi/mid logic.
Week 5-6	Trees (Q176-Q231) + Heaps (Q232-Q255) - DFS/BFS recursion, BST operations, Trie, priority queues.
Week 7-8	Graphs (Q256-Q310) - BFS/DFS, topological sort, Union-Find, Dijkstra, MST.
Week 9-10	Dynamic Programming (Q311-Q390) - start with 1-D, progress to 2-D, knapsack, interval, bitmask.
Week 11	Backtracking (Q391-Q420) + Greedy (Q421-Q445) - pruning strategies, exchange argument proofs.

Week 12

Hashing & Math (Q446-Q470) + Design (Q471-Q500) - mock interviews; time yourself strictly.

Closing Remarks

Consistent, pattern-focused practice is the most reliable path to success in FAANG-level technical interviews. Rather than memorising solutions, internalise the underlying pattern for each problem category. When you encounter an unfamiliar problem, identify which pattern it resembles and apply the corresponding template. Complement this list with timed mock interviews, peer code reviews, and systematic complexity analysis (time and space) for every solution you write. Best of luck with your preparation. You have everything you need - now execute.